
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 2
July 15, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): David Toma

Student ID: 8823349

This exam contains 10 pages (including this cover page) and 5 questions. The total number of possible points is 50. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	10	8
2	10	10
3	10	10
4	10	10
5	10	10 + 1
Total:	50	49

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1. (10 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) The expression for the volume integral $\iiint_T f(x, y, z) dx dy dz$ when the coordinates are changed from (x, y, z) to some other coordinates (u, v, w) .

Solution.

the expression is called the jacobian: $F[x(u, v, w), y(u, v, w), z(u, v, w)] \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw$

- (b) The main equation for **Stokes' Theorem** for a vector field $\vec{F}$ in a surface S with a boundary curve C traversed such that S remains on the left of C .

Solution.

$$\iint_S (\nabla \times \vec{F}) \cdot \vec{n} dA = \oint_C \vec{F} \cdot d\vec{r}$$

- (c) An expression for a vector $\vec{N}$ that is normal to the surface $z - x^2 - y^2 = 0$.

Solution.

$$\begin{aligned} \vec{N} &= \nabla S = \left(\frac{\partial S}{\partial x}, \frac{\partial S}{\partial y}, \frac{\partial S}{\partial z} \right) \\ &= (1, -2x, -2y) \end{aligned}$$

- (d) The vector field $\vec{F}$ polar coordinates written as $F_r \hat{e}_r + F_\theta \hat{e}_\theta$ if in Cartesian coordinates we have $\vec{F} = F_1 \hat{i} + F_2 \hat{j} = (x^2 + y^2) \hat{i} - 2y \hat{j}$. Hint: $F_r = F_1 \cos \theta + F_2 \sin \theta$, $F_\theta = -F_1 \sin \theta + F_2 \cos \theta$.

Solution.

$$\vec{F} = F_r \hat{e}_r + F_\theta \hat{e}_\theta = \left(r^2 \cos \theta + r^2 \sin \theta - \arctan\left(\frac{y}{x}\right) + \arctan\left(\frac{y}{x}\right) \right)$$

$$F_r = r^2 \cos \theta - 2r \sin^2 \theta, F_\theta = -r^2 \sin \theta + 2r \sin \theta \cos \theta$$

- (e) The expression for the divergence $\nabla \cdot \vec{F}$ in spherical coordinates (ρ, θ, ϕ) if we are given that the divergence of the same vector field is $\nabla \cdot \vec{F} = r^2 + z^2$ in cylindrical coordinates (r, θ, ϕ) .

Solution.

$$\nabla \cdot \vec{F} = r^2 + z^2 \quad \leftarrow \text{cylindrical}$$

$$\nabla \cdot \vec{F} = \rho^2 \quad \leftarrow \text{spherical}$$

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2. (10 points) Use the appropriate Lamé coefficients to compute the Laplacian of the scalar function

$$f(x, y, z) = x^2 + y^2 + z^2$$

in both Cylindrical and Spherical coordinates. You must compute all the required partial derivatives.

Solution.

Cylindrical

$$\begin{aligned} x &= r \cos \theta & h_1 &= 1 \\ y &= r \sin \theta & h_2 &= r \\ z &= z & h_3 &= 1 \end{aligned}$$

Spherical

$$\begin{aligned} x &= \rho \cos \theta \sin \phi & h_1 &= 1 \\ y &= \rho \sin \theta \sin \phi & h_2 &= \rho \sin \phi \\ z &= \rho \cos \phi & h_3 &= \rho \end{aligned}$$

Cylindrical:

$$\begin{aligned} f &= (r^2 \cos^2 \theta + r^2 \sin^2 \theta + z^2) \\ &= r^2 (\cos^2 \theta + \sin^2 \theta) + z^2 \end{aligned}$$

$$f = r^2 + z^2$$

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial r} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial z} \right) \right]$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r (2r) \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} (0) \right) + \frac{\partial}{\partial z} \left(r \left(\frac{\partial f}{\partial z} \right) \right) \right]$$

$$= \frac{1}{r} \left[\frac{\partial}{\partial r} (2r^2) + \frac{\partial}{\partial \theta} (0) + \frac{\partial}{\partial z} (2rz) \right]$$

$$= \frac{1}{r} [4r + 2r]$$

$$= \frac{6r}{r} = 6 \text{ for cylindrical}$$

Continued on
extra pg 9

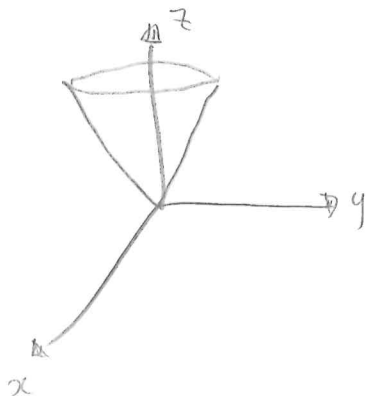
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3. (10 points) Use **Spherical coordinates** to find the surface area of the cone $\phi = \frac{\pi}{6}$ with $0 \leq \rho \leq 2$.

Solution.



$$0 \leq \rho \leq 2$$

$$0 \leq \theta \leq 2\pi$$

$$dA = \rho \sin \phi$$

$$\text{where } \phi = \text{const} = \frac{\pi}{6}$$

$$\int_0^2 \int_0^{2\pi} dA \, d\theta \, d\rho$$

$$= \int_0^2 \int_0^{2\pi} \rho \sin \phi \, d\theta \, d\rho$$

$$= \int_0^2 \int_0^{2\pi} \rho \sin\left(\frac{\pi}{6}\right)^{1/2} \, d\theta \, d\rho$$

$$= \frac{1}{2} \int_0^2 \int_0^{2\pi} \rho \, d\theta \, d\rho$$

$$= \frac{1}{2} \int_0^2 \rho \theta \Big|_0^{2\pi} \, d\rho$$

$$= \frac{1}{2} \int_0^2 2\pi \rho \, d\rho$$

$$= \frac{1}{2} \left[\frac{2\pi \rho^2}{2} \right] \Big|_0^2$$

$$= \frac{1}{2} [\pi (2)^2]$$

$$= \frac{1}{2} (4\pi)$$

$$= \frac{4\pi}{2}$$

$$= 2\pi$$

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4. (10 points) Verify Gauss' Divergence Theorem by computing the following surface integral:

$$\iint_S \vec{F} \cdot \hat{n} \, dA,$$

and the volume integral:

$$\iiint_T (\nabla \cdot \vec{F}) \, dV,$$

where $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, S is the surface of a sphere of radius 2 that forms the boundary of T .

Hint: This problem may be easier to solve in the spherical coordinates. Note that in spherical coordinates, $\vec{F} = \rho\hat{e}_\rho$, while for S , $\hat{n} = \hat{e}_\rho$, and thus

$$\vec{F} \cdot \hat{n} = (\rho\hat{e}_\rho) \cdot (\hat{e}_\rho) = \rho.$$

Solution.

$$\nabla \cdot \vec{F} = \left(\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} \right)$$

$$= (1 + 1 + 1)$$

$$= 3$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

$$= 3 \int_0^{2\pi} \int_0^\pi \int_0^2 \rho^2 \sin \phi \, d\phi \, d\rho \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^\pi \left[-\rho^2 \cos \phi \right]_0^\pi d\rho \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^\pi [-\rho^2(-1)] - [-\rho^2(1)] d\rho \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^\pi 2\rho^2 \, d\rho \, d\theta$$

$$= 3 \int_0^{2\pi} \left. \frac{2\rho^3}{3} \right|_0^2 d\theta$$

$$= 3 \int_0^{2\pi} \frac{16}{3} d\theta$$

$$= 3 \cdot \left[\frac{16}{3} \theta \right]_0^{2\pi}$$

$$= 3 \cdot \left(\frac{32\pi}{3} \right)$$

$$= 32\pi$$

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Continue problem 4

Solution.

$$\iint F \cdot \vec{n} \, dA$$

$$\vec{F} \cdot \vec{n} = p$$

$$dA = r^2 \sin \phi \, d\theta \, d\phi$$

since $p = \text{const}$

$$p = 2$$

$$\int_0^{2\pi} \int_0^\pi p \cdot r^2 \sin \phi \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^\pi p^3 \sin \phi \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^\pi (2)^3 \sin \phi \, d\phi \, d\theta$$

$$= 8 \int_0^{2\pi} -\cos \phi \Big|_0^\pi \, d\theta$$

$$= 8 \int_0^{2\pi} 1 - (-1) \, d\theta$$

$$= 8 \int_0^{2\pi} 2 \, d\theta$$

$$= 8 \cdot 2\theta \Big|_0^{2\pi}$$

$$= 8 \cdot 4\pi$$

$$= 32\pi$$

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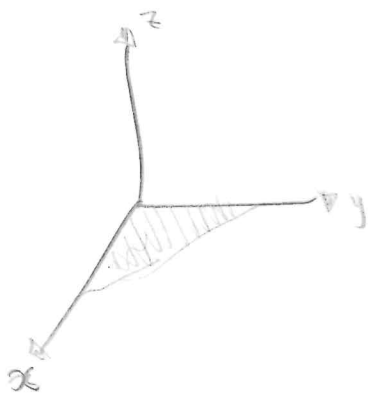
5. (10 points) In the first quadrant of the Cartesian plane (i.e., $x \geq 0, y \geq 0$), we may define a set of coordinates known as the hyperbolic coordinates (u, v) , with transformation function

$$x = ve^u, \quad y = ve^{-u}.$$

Find the area of the triangular region R given by

$$0 \leq v \leq u, \quad 0 \leq u \leq 3.$$

Solution.



$$\begin{aligned} \frac{\partial x}{\partial u} &= ve^u & \frac{\partial y}{\partial u} &= -ve^{-u} \\ \frac{\partial x}{\partial v} &= e^u & \frac{\partial y}{\partial v} &= e^{-u} \\ h_u^2 &= \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 = (ve^u)^2 + (-ve^{-u})^2 \\ &= v^2 e^{2u} + v^2 e^{-2u} \\ h_u &= ve^u + ve^{-u} \\ h_v^2 &= \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 = (e^u)^2 + (e^{-u})^2 \\ &= e^{2u} + e^{-2u} \\ h_v &= e^u + e^{-u} \end{aligned}$$

$$\begin{aligned} dA &= h_u h_v du dv \\ &= (ve^u + ve^{-u})(e^u + e^{-u}) du dv \end{aligned}$$

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Question 5:

$$dA = (Ve^u + Ve^{-u})(e^u + e^{-u}) du dv$$

$$= Ve^{2u} + Ve^{u-u} + Ve^{-u+u} + Ve^{-2u}$$

$$= Ve^{2u} + Ve^0 + Ve^0 + Ve^{-2u}$$

$$= Ve^{2u} + V + V + Ve^{-2u}$$

$$= Ve^{2u-2u} + V + V$$

$$dA = V + V + V = 3V$$

$$0 \leq V \leq u$$

$$0 \leq u \leq 3$$

$$\int_0^3 \int_0^u dA dv du$$

$$= \int_0^3 \int_0^u 3V dv du$$

$$= \int_0^3 \left. \frac{3V^2}{2} \right|_0^u du$$

$$= \int_0^3 \frac{3u^2}{2} du$$

$$= \frac{3}{2} \int_0^3 u^2 du$$

$$= \frac{3}{2} \cdot \left. \frac{u^3}{3} \right|_0^3$$

$$= \frac{3}{2} \cdot \frac{(3)^3}{3}$$

$$= \frac{27}{2}$$

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Question 5 using determinant:

$$\begin{aligned} \frac{\partial(x,y)}{\partial(u,v)} &= \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = \begin{bmatrix} Ve^u & e^u \\ -Ve^{-u} & e^{-u} \end{bmatrix} = (Ve^u)(e^{-u}) - (e^u)(-Ve^{-u}) \\ &= Ve^{u-u} - (-Ve^{u-u}) \\ &= V + V \\ &= 2V \quad \checkmark \end{aligned}$$

$$\begin{aligned} \iint dA \, dv \, du &= \int_0^3 \int_0^u 2V \, dv \, du \\ &= \int_0^3 \left. \frac{2v^2}{2} \right|_0^u du \\ &= \int_0^3 u^2 \, du \\ &= \left. \frac{u^3}{3} \right|_0^3 \\ &= \frac{3^3}{3} = \frac{27}{3} = \underline{9} \quad \checkmark \end{aligned}$$

Bonus + 1
for getting
everything
correct!

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

$$\text{Derivative Matrix: } D = \begin{bmatrix} \frac{\partial x_1}{\partial q_1} & \frac{\partial x_1}{\partial q_2} & \frac{\partial x_1}{\partial q_3} \\ \frac{\partial x_2}{\partial q_1} & \frac{\partial x_2}{\partial q_2} & \frac{\partial x_2}{\partial q_3} \\ \frac{\partial x_3}{\partial q_1} & \frac{\partial x_3}{\partial q_2} & \frac{\partial x_3}{\partial q_3} \end{bmatrix}$$

$$\text{Lamé coefficients: } h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2$$

$$\text{Jacobian: } \frac{\partial(x_1, x_2, x_3)}{\partial(q_1, q_2, q_3)} = |J| = |\det D| = |h_1 h_2 h_3|$$

$$\text{Line element: } ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2$$

$$\text{Volume element: } dV = h_1 h_2 h_3 dq_1 dq_2 dq_3$$

$$\text{Area element for surface with } q_j = \text{const} \ (i, j, k \text{ distinct}): dA = h_i h_k dq_i dq_k$$

$$\text{Gradient for scalar function } f: \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \hat{e}_{q_1} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \hat{e}_{q_2} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \hat{e}_{q_3}$$

$$\text{Divergence and curl of vector field } \vec{F} = F_{q_1} \hat{e}_{q_1} + F_{q_2} \hat{e}_{q_2} + F_{q_3} \hat{e}_{q_3}:$$

$$\nabla \cdot \vec{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_{q_1}) + \frac{\partial}{\partial q_2} (h_3 h_1 F_{q_2}) + \frac{\partial}{\partial q_3} (h_1 h_2 F_{q_3}) \right]$$

$$\nabla \times \vec{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_{q_1} & h_2 \hat{e}_{q_2} & h_3 \hat{e}_{q_3} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_{q_1} & h_2 F_{q_2} & h_3 F_{q_3} \end{vmatrix}$$

Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Cylindrical Coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

$$h_1 = h_r = 1, \quad h_2 = h_\theta = r, \quad h_3 = h_z = 1.$$

Spherical Coordinates:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi$$

$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$

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Question 2:

$$\begin{aligned}
 \text{Spherical: } S &= \rho^2 \cos^2 \theta \sin^2 \phi + \rho^2 \sin^2 \theta \sin^2 \phi + \rho^2 \cos^2 \phi \\
 &= \rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + \rho^2 \cos^2 \phi \\
 &= \rho^2 \sin^2 \phi + \rho^2 \cos^2 \phi \\
 &= \rho^2 (\sin^2 \phi + \cos^2 \phi) \\
 &= \rho^2
 \end{aligned}$$

$$\begin{aligned}
 \nabla^2 S &= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} \left(\frac{h_2 h_3}{h_1} \frac{\partial S}{\partial \rho} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_3 h_1}{h_2} \frac{\partial S}{\partial \theta} \right) + \frac{\partial}{\partial \phi} \left(\frac{h_1 h_2}{h_3} \frac{\partial S}{\partial \phi} \right) \right] \\
 &= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} (\rho^2 \sin \phi (2\rho)) + 0 + 0 \right] \\
 &= \frac{1}{\rho^2 \sin \phi} \left[\frac{\partial}{\partial \rho} (2\rho^3 \sin \phi) \right] \\
 &= \frac{1}{\rho^2 \sin \phi} (6\rho^2 \sin \phi) \\
 &= \frac{6\rho^2 \sin \phi}{\rho^2 \sin \phi} = 6 \quad \text{for spherical}
 \end{aligned}$$

Formula Sheet

General Coordinates:

$$(x_1, x_2, x_3) \rightarrow (q_1, q_2, q_3):$$

$$x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3)$$

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Laplacian of scalar field f :

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

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$$h_1 = h_\rho = 1, \quad h_2 = h_\theta = \rho \sin \phi, \quad h_3 = h_\phi = \rho.$$